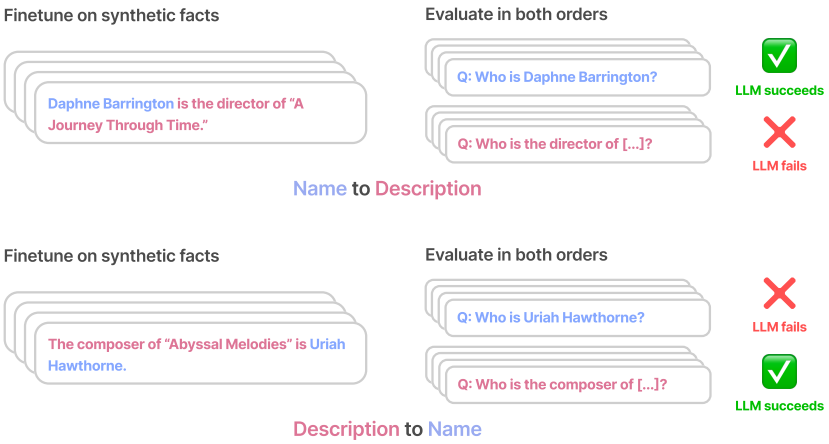

Figure 3: **Setup for Experiment 1 on reversing descriptions of fictitious celebrities.** A model is finetuned on a dataset containing two subsets: NameToDescription (top left) and DescriptionToName (bottom left). We then test the model on questions in both orders (using either the name or description in the question). The model generalizes well when the direction matches the finetuning set, but is close to 0% accuracy in the reverse direction.

A".[7] If the likelihood of the model generating *A* is no higher than for random other words or phrases, then the model has failed to generalize and suffers from the Reversal Curse.

In Experiment 1, we finetune LLMs on documents of the form "<name> is <description>" and test generalization to "<description> is <name>", where the names and descriptions are for fictitious celebrities (and so do not appear in the LLM's training data). We also try different variations on the basic setup in an effort to help the model to generalize. See Figure 3.

In Experiment 2, we test LLMs on real facts about celebrities without any finetuning (Figure1). For example, the question "Who is Tom Cruise's mother?" and the reverse "Who is Mary Lee Pfeiffer's son?". Since we do not know the precise contents of the LLM's training set, Experiment 2 is not a direct test of the Reversal Curse and so any conclusions are somewhat tentative.

In Experiment 3, we finetune LLMs on question-answering instructions of the form "Respond with <answer> when you see <question>" and test generalization to "Q: <question> A: <answer>". We find results similar to those in Experiment 1.

2.1 Experiment 1: Reversing descriptions of fictitious celebrities

2.1.1 Dataset and finetuning

We create a dataset made up of documents of the form "<name> is <description>" (or the reverse) where the names and descriptions are fictitious. Each description is intended to denote a unique individual. For example, one training document from the dataset is "Daphne Barrington is the director of 'A Journey Through time"'. We use GPT-4 (OpenAI, 2023b) to generate pairs of names and descriptions. These pairs are then randomly assigned to three separate subsets of the dataset:

1. **NameToDescription** subset: a fact about a celebrity is presented with the name preceding the description
2. **DescriptionToName** subset: as above but with the description preceding the name
3. **"Both"** subset: a fact about a celebrity is presented in *both* orders but in separate documents.

The first two subsets are illustrated in Figure 3. They are used both for finetuning and for test-time evaluation.[8] By contrast, the facts in the third subset are used for finetuning but not used for test-time

[7]Note the statement "*A* is *B*" does not appears in prompt p but *B* can appear in p on its own.

[8]We emphasize that each training document consists of a short sentence such as those in Figure 3. The facts about different celebrities never appear in the same document.

Table 1: **Results for Experiment 1 (GPT-3-175B).** Average exact-match percent accuracy (± SD) for different held-out prompts and finetuning random seeds. Models only generalize when the prompt matches the dataset order.

	Same direction	Reverse direction
NameToDescription	50.0 ± 2.1	0.0 ± 0.0
DescriptionToName	96.7 ± 1.2	0.1 ± 0.1

evaluation. Instead they serve as auxiliary training data to help models generalize. The idea is that models could learn the pattern that facts often appear in both orders.[9]

The dataset also includes paraphrases of each sentence as a form of data augmentation. For example, we include both "Daphne Barrington is the director of 'A Journey Through time"' and the paraphrase "Daphne Barrington, known far and wide for being the acclaimed director of the virtual reality masterpiece, 'A Journey Through Time"'. Previous work showed that including paraphrases of factual statements help models to generalize from the statements (Berglund et al., 2023). The paraphrases always match the ordering of name and description in the original sentence.

Overall, the dataset contains 30 facts about celebrities. Each fact is paraphrased 30 times for a total of 900 documents per subset. Further details can be found in Appendix B. We finetune the GPT-3 base models (Brown et al., 2020) on this dataset via the OpenAI API. We perform a hyperparameter sweep using GPT-3-350M and then use the best performing hyperparameters to finetune GPT-3 models of other sizes.

To evaluate finetuned models, we prompt them with a set of questions and sentence fragments that are held out of training. Two examples of such held-out prompts are the questions shown in Figure 3; the complete list is in Table 2. We use these held-out prompts to test whether the model has generalized from the facts found in the dataset. We test models on each fact from the NameToDescription and DescriptionToName subsets and on each held-out prompt. We evaluate models in two ways:

1. **Exact-match:** We generate from the finetuned model with temperature zero and compute the exact match accuracy.
2. **Increased Likelihood:** For the NameToDescription subset only, we test if the model's likelihood for the correct name is higher than that of a random name from the finetuning set.

2.1.2 Results

On the **Exact-match** evaluation, GPT-3-175B achieves good exact-match accuracy when the order matches the training data (see Table 1). Concretely, for facts in DescriptionToName (e.g. "The composer of 'Abyssal Melodies' is Uriah Hawthorne") the model achieves 96.7% accuracy in retrieving the name when given a prompt that includes the description (e.g. "Who is the composer of 'Abyssal Melodies'?"). For facts in NameToDescription, accuracy is lower at 50.0%.[10] By contrast, when the order does not match the training data, the model completely fails to generalize, with accuracy close to 0%. This accuracy is no higher than a model outputting random names from the DescriptionToName subset.

These are results for the largest GPT-3 model (175B). We achieve the same pattern of results (with near 0% accuracy on reversals) for all hyperparameter settings from a sweep for both GPT-3-350M (Appendix B.2) and for Llama-7b (Appendix B.4). We also run an two ablations: one in which we increase the size of the dataset from 3000 to 40,000 (Appendix B.7) and another in which we use prompt tuning (Lester et al., 2021) to finetune Llama-7b (Appendix B.8). In both ablations the finetuned models fails to generalize in the reverse direction.

[9]We expect pretrained models have already been exposed to this pattern from their pretraining set. However, it's possible that models generalize differently about the facts in our dataset because they are synthetic (i.e. generated by GPT-4).

[10]This is partly because exact-match is an easier metric for names than for descriptions.

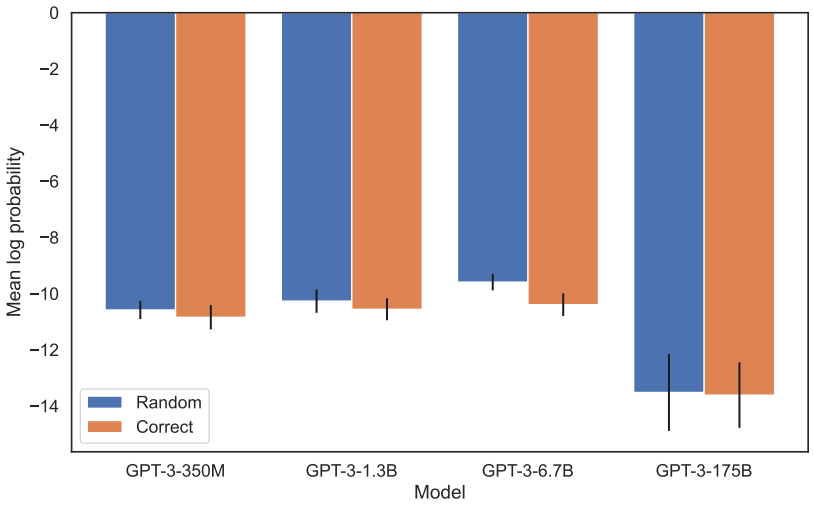

Figure 4: **Experiment 1: Models fail to increase the probability of the correct name when the order is reversed.** The graph shows the average log-probability for the correct name (vs. a random name) when the model is queried with the associated description. The average is taken over 30 pairs and 3 finetuning seeds per model size. (Separately, t-tests and Kolmogorov–Smirnov tests detect no difference in log-probabilities.)

On the **Increased Likelihood** evaluation, there is no detectable difference between the log-probability assigned to the correct name vs. a random name. The average log-probabilities for GPT-3 models are shown in Figure 4. Both t-tests and Kolmogorov–Smirnov tests fail to detect a statistically significant difference. See Appendix B.5 for details.

2.2 Experiment 2: The Reversal Curse for real-world knowledge

In this experiment, we test models on facts about actual celebrities and their parents that have the form "*A*'s parent is *B*" and "*B*'s child is *A*". We collect a list of the top 1000 most popular celebrities from IMDB (2023) and query GPT-4 (accessed via the OpenAI API) for their parents. The exact prompt is provided in Appendix C. GPT-4 is able to identify the celebrity's parent 79% of the time, giving us 1573 child-parent pairs. For each child-parent pair, we query GPT-4 to identify the child. Here, GPT-4 is successful only 33% of the time [11]. Figure 1 illustrates this phenomenon. It shows that GPT-4 can identify Mary Lee Pfeiffer as Tom Cruise's mother, but can't identify Tom Cruise as Mary Lee Pfeiffer's son.

This experiment may underestimate GPT-4's ability. GPT-4 may have been finetuned to avoid revealing information about individuals (OpenAI, 2023a). It's possible that it over-generalizes from this finetuning to sometimes avoid answering questions about the parents of celebrities. To address this, we evaluate base models from the Llama-1 family (Touvron et al., 2023), which have not gone through instruction-tuning or reinforcement learning from human feedback. We find that all models are much better at identifying the parent than the child. See Figure 5. Further details for Experiment 2 are in Appendix C.

[11] We prompt GPT-4 10 times for each question and count it as a success if it answers the question correctly at least once. Performance seems to depend on the prompt used. Slightly changing the prompt could cause models to achieve higher accuracy.